Reading Arrays: Rows, Groups, and the Total
Answer Key

Grade 2–3 Multiplication

Quick Answers

1. 15	3. 18	5. 16	7. 6
2. 24	4. 20	6. 7	8. 4

Curriculum

Level: Grade 2–3 Multiplication
3.OA.A.3 / 4.OA.A.2 – *problems 4, 5, 6, 7, 8*
3.OA.C.7 – *problems 1, 2, 3*
Difficulty 1–4 of 5 across 8 tagged checks.

Common wrong answers

2. *If they got 4: wrote down the number of rows instead of the total number of counters*
3. *If they got 9: added the number of rows to the number in each row instead of multiplying*
4. *If they got 5: reported the number of rows as the total number of stickers*
5. *If they got 10: added the rows and the number in each row instead of multiplying*

1. Array of dots: 3 rows, 5 in each row. How many in all?

Solution: Name the two numbers before multiplying. The number of *rows* is 3 (count down the left side). The number *in each row* is 5 (count across one row). The question asks for the total, so multiply rows × in each row: 3×5. Skip-count by fives three times — 5, 10, 15 — or add $5 + 5 + 5$. Total dots: $\boxed{15}$.

2. Array of squares: 4 rows, 6 in each row. Write the sentence and find the total.

Solution: The multiplication sentence records the picture in order: rows first, then how many are in each row, so it is $4 \times 6 = \square$. Skip-count by sixes four times: 6, 12, 18, 24. Total squares: $\boxed{24}$.
Watch for: a child who answers 4 has reported the number of rows, which is the count of *groups*, not the count of *squares*.

3. Garden: 6 rows of carrots, 3 carrots in each row.

Solution: The words "in the garden altogether" name the total, so the answer must count carrots, not rows. Rows = 6, in each row = 3, so the total is 6×3. Skip-count by threes six times: 3, 6, 9, 12, 15, 18. Carrots in all: $\boxed{18}$.
Watch for: the answer 9 comes from $6 + 3$. Adding joins two groups; here there are six equal groups, which is multiplication.

4. Find and fix: Jonah's 5 rows of 4 stickers.

Solution: Jonah answered the question "how many rows?" when the question asked "how many stickers in all?" The 5 he counted is the number of groups; each of those groups still holds 4 stickers. Multiply the two numbers he already found: 5×4. Skip-count by fours five times: 4, 8, 12, 16, 20. Correct total: $\boxed{20}$.

5. Find and fix: Mia's 2 rows of 8 shells.

Solution: Mia used addition, which would be right only if she had one group of 2 and one group of 8. She actually has 2 equal groups of 8, and equal groups call for multiplication: 2×8. Doubling 8 gives 16, and the picture agrees — $8 + 8 = 16$. Correct total: $\boxed{16}$.
Watch for: her answer 10 is $2 + 8$; it is smaller than one full row of shells, which is a quick signal that the operation was wrong.

6. 28 muffins in 4 equal rows — how many in each row?

Solution: Here the *total* is known (28) and the number of rows is known (4); the missing number is the size of one group. The multiplication sentence is $4 \times n = 28$. Ask "4 times what is 28?" — skip-count by fours: 4, 8, 12, 16, 20, 24, 28, which took 7 counts. Check: $4 \times 7 = 28$ ✓. Muffins in each row: $\boxed{n = 7}$.

7. 30 beads in rows of 5 — how many rows?

Solution: This time the size of each group is known (5 beads) and the number of rows is missing: $5 \times n = 30$. Skip-count by fives until reaching 30: 5, 10, 15, 20, 25, 30 — that is 6 counts, so 6 rows. Check: $5 \times 6 = 30$ ✓. Number of rows: $\boxed{n = 6}$.

8. Challenge — 36 counters in rows of 9.

Solution: Sam read the total (36 counters) as if it were the number of rows. Those are different quantities: 36 counts every counter on the board, while a row is a whole group of 9 counters. The sentence for the board is $9 \times n = 36$, where n is the number of rows. Skip-count by nines: 9, 18, 27, 36 — four counts. Check: $9 \times 4 = 36$ ✓. Number of rows: $\boxed{n = 4}$.